

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

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Abstract

Large pre-trained language models have been shown to store factual knowledge in their parameters, and achieve state-of-the-art results when fine-tuned on downstream NLP tasks. However, their ability to access and precisely manipulate knowledge is still limited, and hence on knowledge-intensive tasks, their performance lags behind task-specific architectures. Additionally, providing provenance for their decisions and updating their world knowledge remain open research problems. We introduce a general-purpose fine-tuning recipe for retrieval-augmented generation (RAG) models which combine pre-trained parametric and non-parametric memory for language generation.

1. Introduction

Pre-trained neural language models have been shown to learn a substantial amount of in-depth knowledge from data. They can do so without any access to an external memory source, as a parameterized implicit knowledge base.

While this development is very exciting, such models do have downsides: they cannot easily expand or revise their memory, can't straightforwardly provide insight into their predictions, and may produce hallucinations.

We propose retrieval-augmented generation (RAG), which addresses these issues by combining a pre-trained seq2seq model with a dense passage retriever over a non-parametric memory such as Wikipedia.

2. Method

Our RAG architecture uses a pre-trained seq2seq transformer as the parametric memory and a dense vector index of Wikipedia as the non-parametric memory.

The input sequence x is used to retrieve text documents z , which are then used as additional context when generating the target sequence y .

We consider two variants: RAG-Sequence, which uses the same retrieved document to generate the complete sequence, and RAG-Token, which can use different documents for each generated token.

The retriever is a dense passage retriever (DPR) with a bi-encoder architecture, and the generator is a BART-large sequence-to-sequence model.

3. Experiments

We evaluate RAG on three open-domain question answering tasks: Natural Questions, TriviaQA, and WebQuestions.

RAG-Sequence and RAG-Token both achieve state-of-the-art results on all three datasets, outperforming both closed-book models and retrieve-and-extract baselines.

Ablation studies confirm that the non-parametric memory is critical: removing retrieval leads to a substantial drop in accuracy.

4. Limitations

The retrieval step introduces additional computational cost at inference time, which limits deployment in latency-sensitive settings.

Our approach depends on the quality and coverage of the external corpus; when relevant documents are absent, the model still hallucinates.

End-to-end training of the retriever and generator remains expensive, requiring significant GPU resources.